Class 40: Physics of Nano-Scale Materials

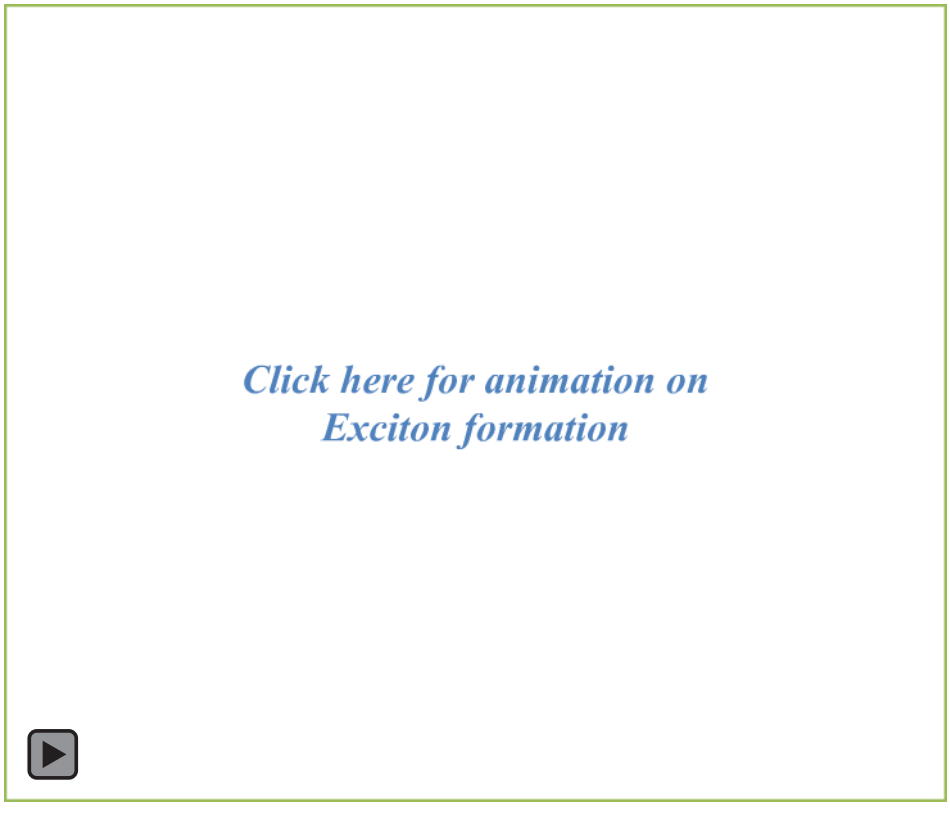
Nano-scale materials, or nanomaterials, are materials where the particle or crystal size is in the scale of a few nm. In recent years there has been considerable interest in the area of nanomaterials in view of the interesting ways in which their properties can be manipulated. In general, most material properties change as a function of size. The trend in the size dependence of a property may be different in the macroscopic state, when compared to that in the nano regime. Trends in the size dependence of properties may get exaggerated or even reversed when the size scale decreases to a few nano meters. It is therefore of interest to examine nanomaterials since it becomes possible to obtain vastly different properties while the chemical composition is the same.

It is necessary to note, that while the particle size or crystal size being a few nm, in some ways qualifies the material as a nanomaterial, for each property and each material, interesting size dependence of properties may become apparent only below a specific length. In other words, there is no universal size below which all nanomaterials show interesting trends in properties. In some cases the properties may become interesting below a few tens of nm, and in some other cases the properties may become interesting only when the size drops even further to below a few nm. So if the material simply has a particle size or crystal size of a few nm, but its properties are exactly the same as that of a bulk mater, then there is no interest in this nanomaterial.

In this class we will look at the impact of nano scale materials on their electrical and optical properties.

Let us first examine what is the size scale that is of interest for nanomaterials to show electrical and optical properties that are different from the bulk properties of the same material. We have noted earlier that in intrinsic semiconducting materials, optical absorption occurs by transition of an electron from the valence band to the conduction band. This transfer of an electron to the conduction band creates a hole in the valence band, as shown in Figure 40.1 below.

Animation of figure 40.1: Exciton formation



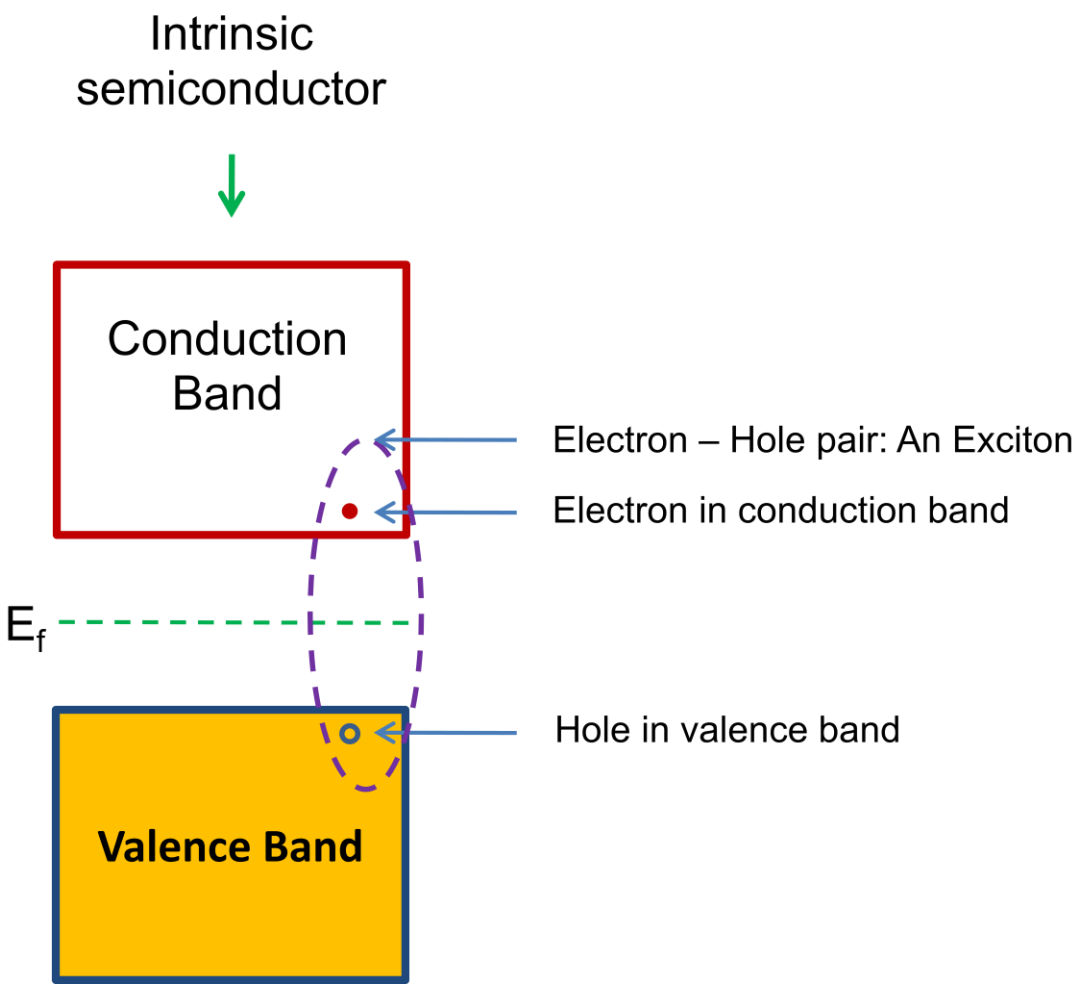


Figure 40.1: Transfer of an electron to the conduction band and the resultant formation of a hole in the valence band of an intrinsic semiconductor, due to the absorption of a photon of appropriate energy. The electron-hole pair exert attractive forces on each other and behave like a combined entity, called the Exciton, which has some similarities to the hydrogen atom in view of the number of particles involved and their charges.

Since the electron and hole have opposite charge, they exert an attractive force on each other and maintain a connectivity with each other. This electron hole pair, that operates as an associated pair of particles, is called an 'Exciton'. The attractive force between them makes them more stable than they would be as independent particles.

The exciton has similarities to a hydrogen atom in the sense that it contains two entities of opposite charge, just like the hydrogen atom. Therefore the exciton can be treated similar to a hydrogen atom, and just like the Bohr radius for a hydrogen atom, it is possible to define an exciton Bohr radius. While the Bohr radius of the hydrogen atom is of the order of an angstrom, the exciton Bohr radius can be several nanometers – in this respect the exciton differs from the hydrogen atom. The exact value of the exciton Bohr radius varies from material to material and is dependent on the dielectric constant of the material which impacts the interaction to some degree.

For nanomaterials selected for their optical or electronic properties, the size scale of interest is therefore the exciton Bohr radius for that specific material. This value could vary from system to system and could be specific values such as 3 nm or 8 nm etc. If the nanomaterial is synthesized in this size scale, we begin to see nano scale effects in the electrical and optical properties of the material.

'Confinement' of the exciton, by making the particle size smaller, impacts the band gap displayed by the material. By making the size of the particles smaller, and making the confinement more severe, the band gap increases. By synthesizing particles with a narrow size distribution, but with size in the nanometer range, it is possible to 'tune' the band gap of the material. It is therefore experimentally possible to get several samples of the same material such that the band gap is different in each sample. This is particularly useful for creating devices without having to use dissimilar materials, thereby eliminating diffusion, corrosion, and sealing issues associated with dissimilar material contact.

With regard to synthesizing materials in different size scales, we have a few specific possibilities as summarized in Figure 40.2:

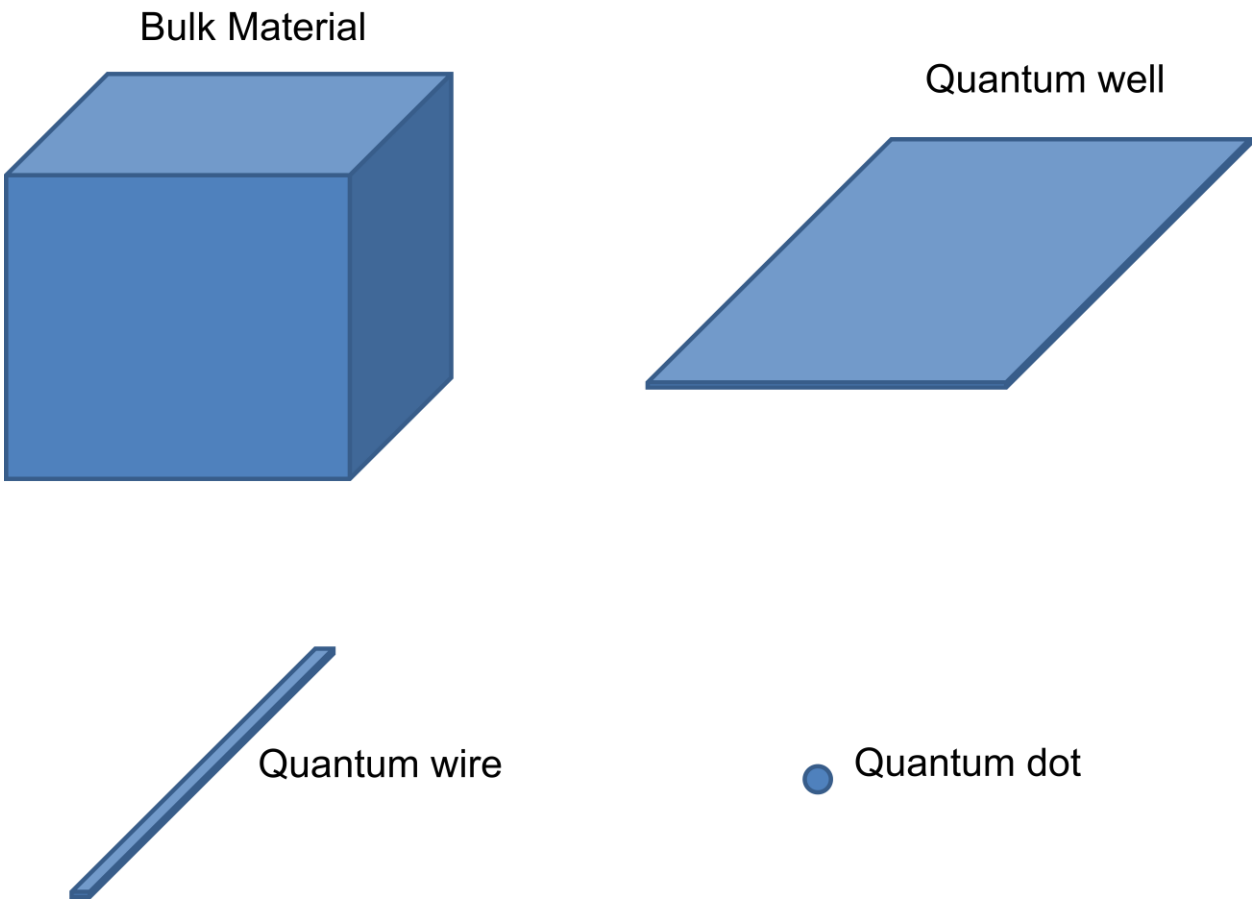
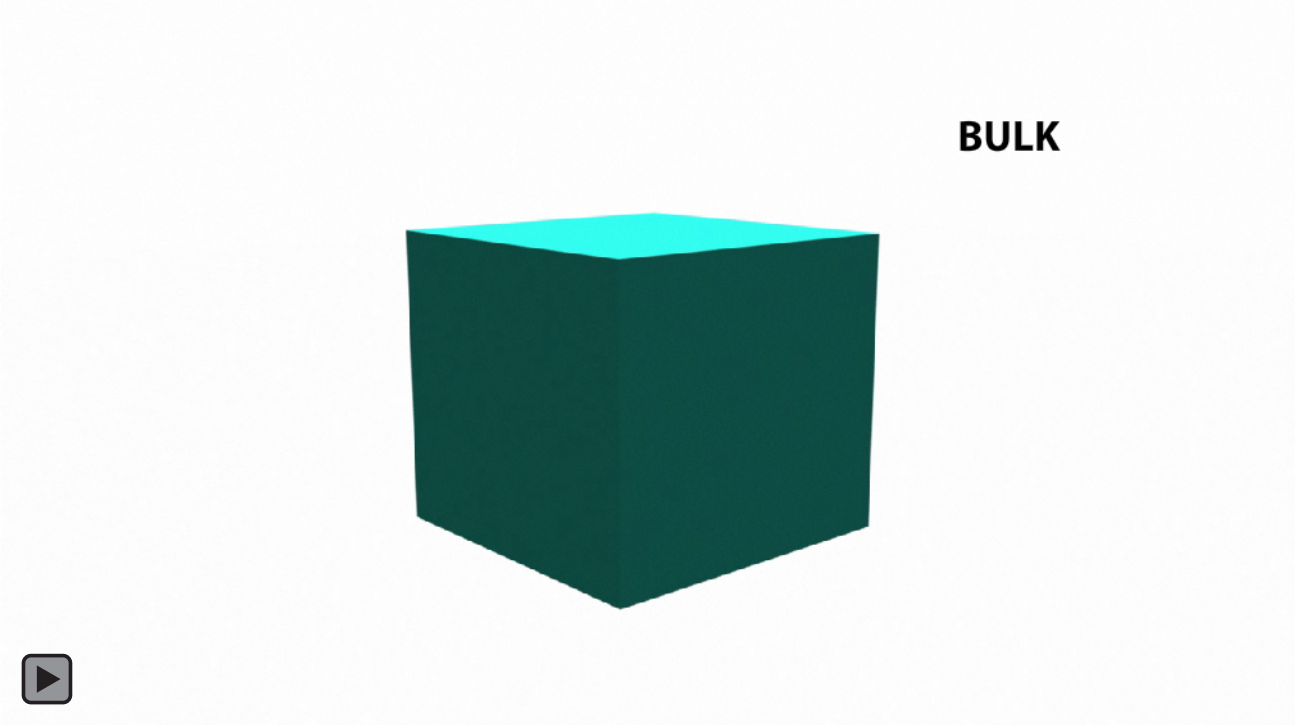


Figure 40.2: Schematic showing a bulk material, a quantum well, a quantum wire and a quantum dot.

Animation of figure 40.2



The possibilities with respect to synthesizing materials in different size scales are:

1) Bulk material. No confinement effects observed.

2) Quantum well: In this case a very thin layer of the material is synthesized, with the thickness of the layer in the range of the exciton Bohr radius. This implies that the exciton is confined in one dimension and free in two other dimensions.

3) Quantum wire: A very narrow wire of the material qualifies as a quantum wire, where the confinement effects are observed in two dimensions, but the exciton is free in one dimension.

4) Quantum dot: The material has a very small size in all three dimensions, and the exciton is therefore confined in all three dimensions.

Materials synthesized in the nano-scale have very high surface area and are therefore very reactive with their environments. It is therefore a challenge to stabilize these particles. Such stability is usually brought about by trapping the material in host matrices such as polymers. The particles can be made to grow within the structure of the polymer and by selecting the polymer and processing conditions, specific narrow size range, nano particles can be synthesized.

Examples where these nanoscale effects have been effectively demonstrated are in the CdS system, and in the PbS system. The bulk band gap of CdS system is in the visible region of the electromagnetic spectrum. By confinement effects the band gap can be tailored or tuned to increase into the UV range of the spectrum. Similarly, the bulk band gap of the PbS system is in the IR region of the visible spectrum, and by introducing confinement effects, the band gap can increase into the visible region of the spectrum and further increase into the UV region of the spectrum.

On an aesthetic note, the above ability to manipulate the band gap implies that with the same chemical composition it is possible to obtain samples with a wide range of colors, across the entire visible spectrum.

A technologically important application of this nano-scale effect on the optical properties of the material is that it becomes possible to synthesize a variety of different materials that can be used for solar cells. Solar radiation has a significant amount of energy in the IR region of the spectrum. Therefore having materials that absorb significantly in this wavelength region of the electromagnetic spectrum, makes them excellent for use in developing solar cells. By utilizing band gap tuning discussed here, it becomes possible to have a larger variety of materials capable of being used for solar cell application, than is possible by merely looking at the bulk band gap of the materials.